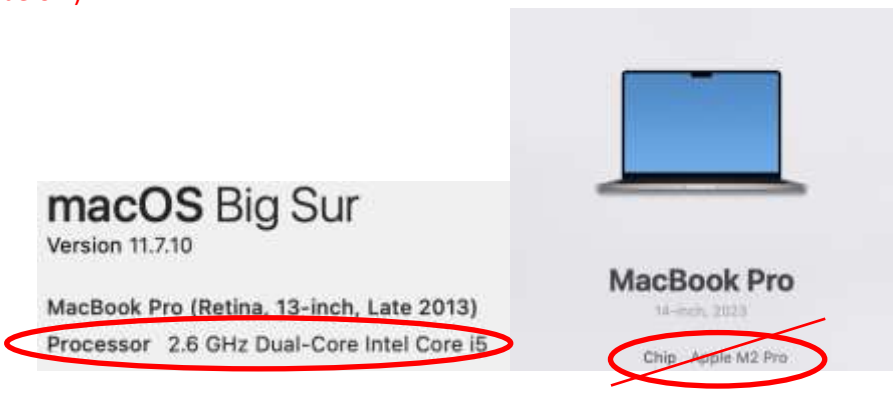
VirtualBox - Dual Boot Virtual Machine - Windows 10 and Ubuntu Linux Installation Guide

1. Overview
   1.1. This document provides a guide for performing the following tasks:
      1.1.1. Install VirtualBox (VB) on your personal PC/laptop using this document as a guide.
      1.1.2. Create a New Virtual Machine (VM) in Virtual Box.
      1.1.3. Install Windows 10 in the new Virtual Machine.
      1.1.4. Install Ubuntu Linux to the same VM and configure to dual-boot.

   1.2. You need approx. 30GB of free space on your laptop's hard drive – NOT a 'cloud' folder.

   1.3. Students must be familiar with the operation of your own computer/laptop to complete these instructions.

   1.4. These instructions are for Windows computers/laptops and MacOSX Intel processor chip laptops only.
      1.4.1. These instructions are NOT applicable for MacOSX laptops using the M1 or M2 processor chip. (See screenshots below).



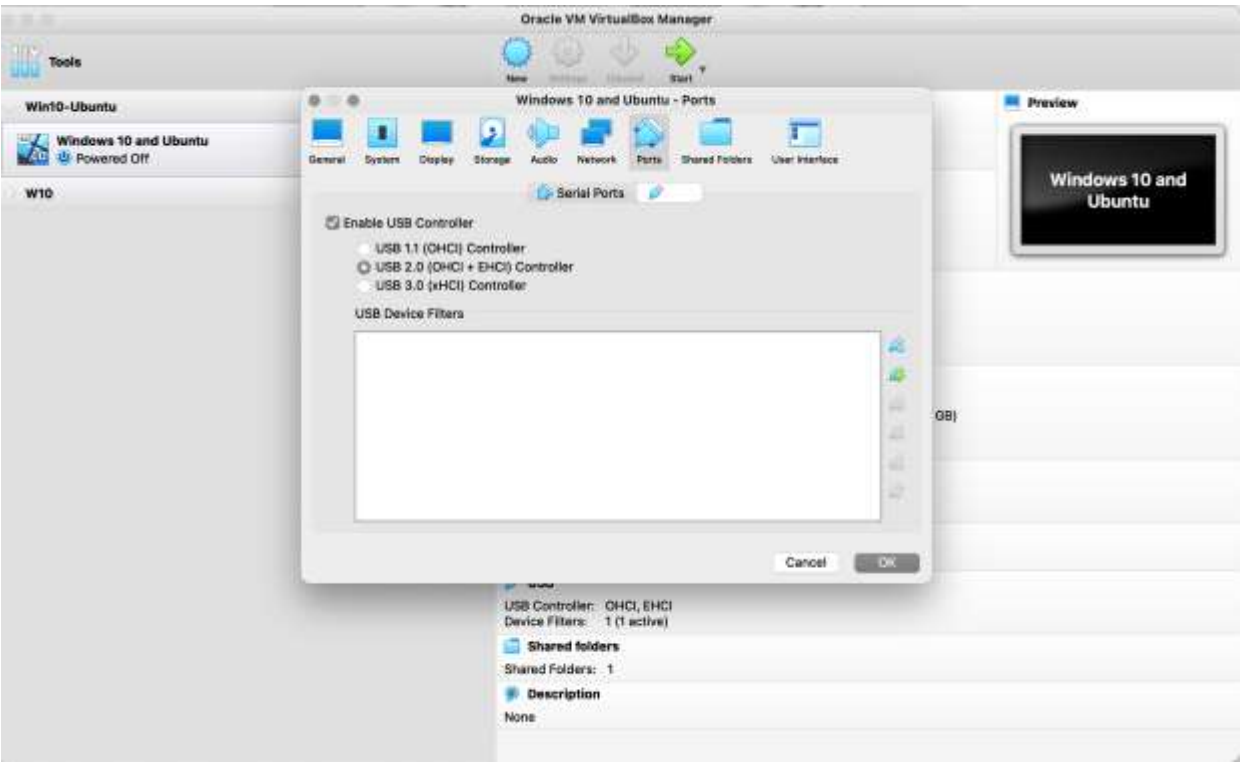
   1.5. If you have no access to a Windows or MacOSX Intel chip computer/laptop, or your Windows or MacOSX laptop is malfunctioning, or you are unable to configure your own laptop, then login to a classroom computer in BB121 or BB123 with netID/Pwd to install and use the required VB-VM during the course.
      1.5.1. Access to classroom computers is based on classroom availability, therefore, do not wait for the last minute to complete the coursework since classroom availability may change.
      1.5.2. **NOTE**: It is recommended that you install VB-VM on a computer in each classroom so you have a back-up to perform your work should a classroom become unavailable.

   1.6. Failure to install VirtualBox and the dual-boot Virtual Machine as outlined in this document will result in substantial grade deductions throughout course.

2. Virtual Box Installation
   2.1. If you do not have VirtualBox already installed in your laptop, then download and install the appropriate ***VirtualBox (VB)*** (Windows or MacOSX) and its corresponding ***VirtualBox Extension Pack*** available via the *VirtualBox web site*.
      2.1.1. Ensure you install VB to your local hard drive of your computer/laptop's Desktop folder and ensure it is NOT installed to the 'Cloud' (OneDrive, etc.), to minimize risk of errors when performing the required coursework.

   2.2. If you already have VB and its corresponding Extension Pack installed in your laptop then proceed with the next step below.

   2.3. After installation of VB and Expansion Pack, configure the VirtualBox Manager settings for usage of the USB drive throughout the course.
      2.3.1. Go to Settings – Ports – USB and enable USB 2.0 controller. (see screenshot)



3. Virtual Machine Installation
   3.1. Download the ***Windows_10_multi-edition_version_x64_100406711.ISO*** via the *BrightSpace Course Materials link* and save file to your computer/laptop.

   3.2. Run VirtualBox and Create a *New* Virtual Machine.

   3.3. Select **30 GB** size of Virtual Disk – **VMDK** file type – **Fixed** Size – Save VM to your laptop and NOT to a cloud folder.

   3.4. Go to Settings - *Storage - Storage Devices/Highlight Empty Disk Icon (sata port 1)* → *Attributes/Optical Drive* → Select the downloaded ***windows_10_multi-edition_version_1709_updated__dec_2017_dvd_100406711.ISO***.

   3.5. Once the settings have been modified, Save and *Start* the *New* Virtual Machine.

   3.6. VM should boot directly from the downloaded Windows **ISO** file to start the Windows Setup and installation of *Windows 10 Pro* to the New Virtual Machine.
      3.6.1. **NOTE**: It is recommended that you change the *VirtualBox* settings by selecting *View/Scaled Mode* from the tool/finder bar and then adjusting the size of the VM windows.

   3.7. When prompted to *Activate Windows* select, '*I don't have a product key*.'

   3.8. When prompted, select **Windows 10 Pro** operating system.

   3.9. When prompted, to select the *Type of Installation* select *Custom: Install Windows Only*.

   3.10. When prompted to 'Where do you want to install Windows?'
      3.10.1. Select '*New'* partition with a size of **22 GB** only (approx. 22,000 MB).
         3.10.1.1. DO NOT USE THE ENTIRE **30GB** allocated to the VM for Windows.
      3.10.2. **NOTE**: If you dedicate the entire **30GB** of space for Windows then you will not be able to install Ubuntu Linux later and will require for you to re-start the Windows installation again.

3.11. The Windows installation process will automatically create two partitions within the **22GB** partition.
- 3.11.1.1. Partition 1 Type: System (System Reserve)
- 3.11.1.2. Partition 2 Type: Primary (Windows installation)
- 3.11.1.3. Unallocated Space (to be used later)

3.12. The installation process will restart VM on its own – do NOT interact with the VM when the installation automatically restarts.

3.13. When prompted select 'Set up for personal use' - '*Offline Account*' - '*Limited Experience*.'

3.14. Create a new user and you must enter **YourLastName SU25** as the primary username.
- 3.14.1. **NOTE**: **YourLastName** means your own last name ONLY (NOT your first name, or first/last name).
- 3.14.2. You may enter a username password or not – it is not needed for this course.

3.15. If a dual-boot virtual machine was previously created, then make sure you create the required username above.

3.16. When prompted you may mute/discontinue 'Cortana' and turn OFF all privacy settings.

3.17. Once *Windows 10 Pro* installation is completed, login to Windows with **YourLastName SU25** username to verify your installation and username was successfully created.

3.18. **NOTE**: Failure to create the usernames listed in this document will result in point deductions throughout the entire course.

3.19. While logged with your **YourLastName SU25** username, create two new Windows users listed below via *Windows Administrative Tools/Computer Management/Local Users and Groups/Users,* as follows:
- 3.19.1. **YourLastName SU25 User 2** and **YourLastName SU25 User 3**.
- 3.19.2. No password necessary for the users – **unselect** *User Must Change Password* for both users.

3.20. **NOTE:** Failure to create the usernames listed in this document will result in point deductions throughout the entire course.

3.21. After creating the additional Windows usernames, then sign out from your primary username and sign into each of the additional user accounts (User 2 and User 3) to verify they were successfully created.
- 3.21.1. **NOTE**: Failure to verify the additional usernames will result in point deductions throughout the entire course.

3.22. Log-off all Windows user accounts and take a **screenshot** of the Windows Sign-in screen and showing the three created Windows user accounts, as follows:
- 3.22.1. **YourLastName SU25**
- 3.22.2. **YourLastName SU25 User 2**
- 3.22.3. **YourLastName SU25 User 3**
- 3.22.4. **NOTE**: Ensure screenshot includes your computer/laptop's taskbar/finder bar showing the date/time. (refer to the *Appendix* for screenshot example)

3.23. Shutdown windows by selecting the Windows VM 'Power' and 'Shutdown.'
- 3.23.1. **NOTE**: Do not shut down the VM via the VirtualBox 'Save the Machine state', 'Send a Shutdown signal' or 'Power off machine' options since it may corrupt the newly installed Windows operating system and you may have to reinstall Windows.

3.24. After Windows shutdown and returning to the VirtualBox Manager remove the ***windows_10_multi-edition_version_1709_updated__dec_2017_dvd_100406711.ISO*** shown in *SATA Port 1*.
- 3.24.1. NOTE: Never remove the **VMDK** file from SATA Port 0.

4. **Ubuntu Linux Installation**
   4.1. Download the ***ubuntu-14.04.1-desktop-amd64.ISO*** via the *BrightSpace Course Materials* link and save to your laptop.

   4.2. Highlight the installed *Windows 10 Pro* VM and go Settings - *Storage - Storage Devices/Highlight Empty Disk Icon → Attributes/Optical Drive* (sata port 1) → Select the **ubuntu-14.04.1-desktop-amd64.ISO** file.

   4.3. Once the settings have been modified, Save and *Start* the VM to start Ubuntu Linux.

   4.4. When prompted by Ubuntu select the install Ubuntu.

   4.5. When prompted select the Installation Type – **Something Else**.
      4.5.1. **NOTE**: DO NOT select 'erase disk and install Ubuntu' since this will erase your Windows partition.
      4.5.2. Highlight Free Space – approx. **10GB**.
      4.5.3. Add partition (**+**) – allocate **8GB** (8000 MB) and only select Use as: **ext4** partition and Mount point (**/**)
         4.5.3.1. **NOTE**: Do not make any changes to any other options.
      4.5.4. Highlight Free Space – approx. **1GB**.
      4.5.5. Add partition (**+**) – allocate the remaining free space size and select Use as: **swap area**.

   4.6. When prompted Create Username: **yourlastname25su** (all lower case) and password.
      4.6.1. If you have a long last name, then you may abbreviate the name, but you must include **'25su'** in the username.
   4.7. **NOTE**: Failure to create the usernames listed in this document will result in point deductions throughout the entire course.

   4.8. Upon completion of Ubuntu installation, it should restart the VM or provide an option to '*Restart Now*' and hit 'Enter' when it stops at the prompt '*Please remove installation media and close the tray (if any).*'

   4.9. **NOTE**: In rare instances, the Ubuntu installation *progress bar* may get 'hung up' for several minutes at the end of the installation process. If this occurs, then shutdown the Ubuntu VM by using the *VirtualBox* 'Power off the machine' option to complete the Ubuntu Installation.

   4.10. After Ubuntu shutdown and returning to the VirtualBox Manager, make sure you remove the **ubuntu-14.04.1-desktop-amd64.ISO** file from *SATA Port 1*.
      4.10.1. **NOTE**: Failure to remove the **ubuntu-14.04.1-desktop-amd64.ISO** file from *SATA Port 1* will cause you to enter the Ubuntu Installation Setup screen every time you start the VM instead of entering the Ubuntu dual-boot menu.
      4.10.2. NOTE: Never remove the **VMDK** file from SATA Port 0.

   4.11. Start the dual-boot VM without any ISO files in the *SATA Port 1* to boot to the Ubuntu dual boot menu choice of *Ubuntu* or the *Windows Recovery Environment* (previously installed Windows 10 VM).

   4.12. Highlight **Ubuntu** and 'Enter' access and login to your primary Ubuntu user **yourlastname25su** to verify the Ubuntu linux installation and username creation were successful.
      4.12.1. If a dual-boot virtual machine was created for with a different username, then make sure you create the required users for this course.
      4.12.2. **NOTE**: Failure to create the usernames listed in this document will result in point deductions throughout the entire course.

      4.12.3. When prompted decline to upgrade Ubuntu.

4.13. While logged in your Ubuntu primary username, create two additional Ubuntu linux users via the *System Settings/User Accounts* or *terminal* commands.

4.14. Create two Ubuntu usernames - **yourlastname25su2** and **yourlastname25su3**.

4.15. If you have a long last name, then you may abbreviate the name, but you must include '**25su'** in the usernames.

4.15.1. Activate both user accounts without a password.

4.15.2. **NOTE**: Failure to create the usernames listed in this document will result in point deductions throughout the entire course.

4.16. If you encounter the Ubuntu display to be too large to navigate to the *User Accounts* settings, then you may add the two additional Ubuntu users using *terminal* commands, as follows:

4.16.1. Enter in the 'Search' icon located in the upper left bar 'User' to access the *User Account settings* and unlocking the primary user account to add the two users, OR

4.16.2. Enter in the 'Search' icon '*Terminal'* to open a command terminal window to type the commands listed in the screenshot below to add both user accounts with the appropriate configuration (see screenshot).

4.17. After creating both Ubuntu user accounts, log off your primary Ubuntu user account.

4.18. Login to each of the two newly created Ubuntu usernames (**yourlastname25su2** and **25su3**) to verify they were properly created.

4.18.1. **NOTE:** Failure to create the usernames listed in this document will result in point deductions throughout the course.

4.19. Logoff all Ubuntu user accounts and take a **screenshot** of the Ubuntu login screen and show the three created Ubuntu user accounts:

4.19.1. **yourlastname25su**

4.19.2. **yourlastname25su2**

4.19.3. **yourlastname25su3**

4.19.4. **NOTE**: Ensure screenshot includes your computer/laptop's taskbar/finder bar showing the date/time. (refer to *Appendix* for screenshot example)

4.20. Perform a normal Ubuntu shutdown via the *Settings* icon on the upper right-hand side of Ubuntu login menu screen.

4.20.1. **NOTE**: Do not shutdown the Ubuntu VM using the *VirtualBox* 'Save the Machine state', 'Send a Shutdown signal' or 'Power off machine' since it may corrupt the newly installed Ubuntu Linux operating system and you may have to reinstall Ubuntu.

4.21. If necessary, after shutting down Ubuntu and returning to the *VirtualBox Manager* remove the ***ubuntu-14.04.1-desktop-amd64.ISO*** file in *SATA Port 1,* if shown.

4.21.1. **NOTE**: Failure to remove the Ubuntu ISO from *SATA Port 1* may cause you to enter the Ubuntu Installation Setup screen every time you start the VM instead of entering Ubuntu you installed with the three Ubuntu usernames.

4.21.2. **NOTE**: Never remove the **VMDK** file from *SATA Port* 0.

5. Ubuntu Bootloader

5.1. Start the dual boot VM and take a **screenshot** of the Ubuntu boot loader menu showing the available operating system selections - *Ubuntu* and *Windows Recovery Environment*.

5.1.1. **NOTE**: Ensure screenshot includes your computer/laptop's taskbar/finder bar showing the date/time. (refer to the *Appendix* for example of the required screenshot)

6. VirtualBox Manager

6.1. Take a **screenshot** of the *VirtualBox Manager* showing all the settings for the created dual-boot virtual machine to show completion of the required installation tasks.

6.1.1. **NOTE**: Ensure screenshot includes your computer/laptop's taskbar/finder bar showing the date/time. (refer to the *Appendix* for example of the required screenshot)

7. Screenshot Submissions

7.1. Submit the required four screenshots via *BrightSpace*, as directed by the instructor.

8. Additional Configuration

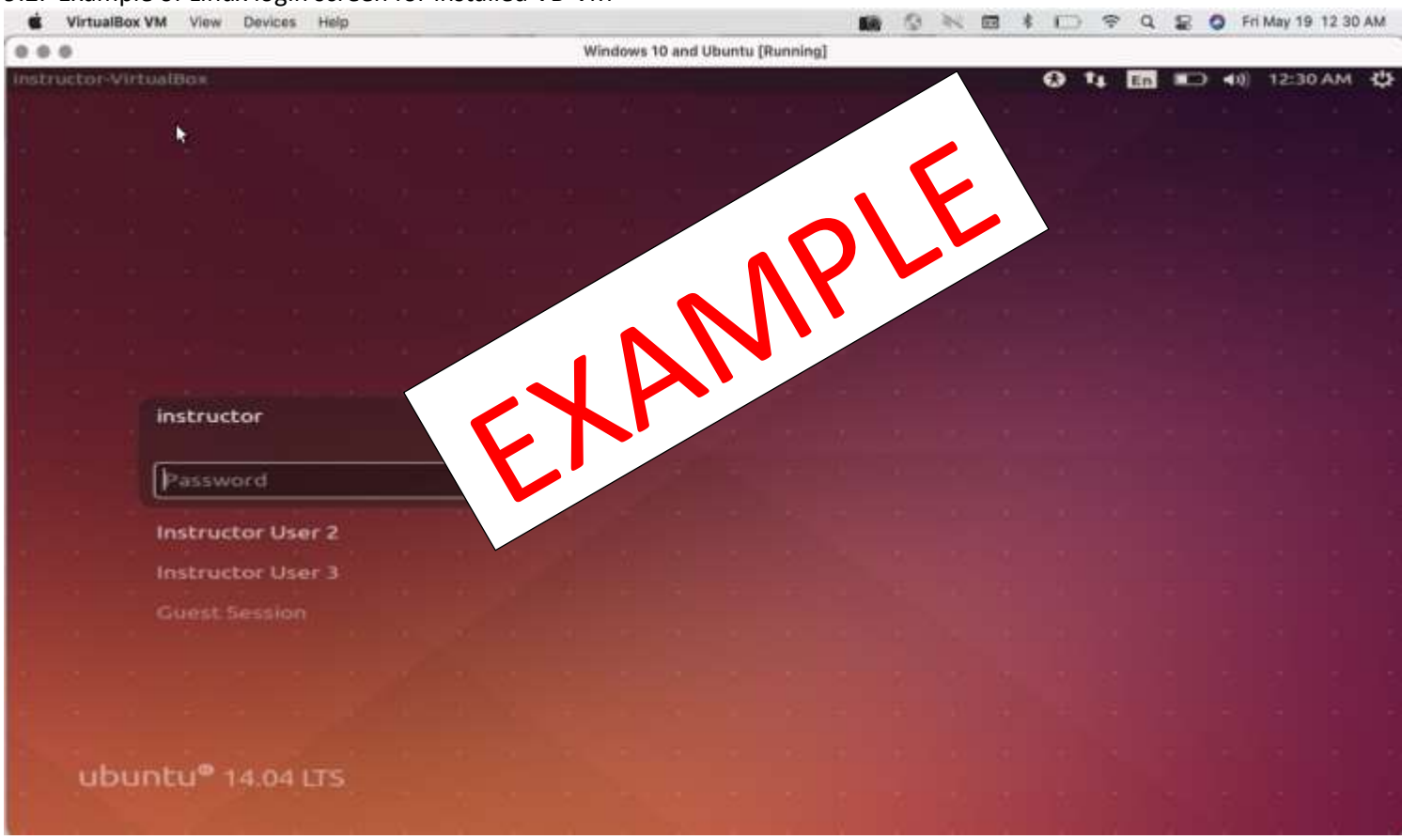
8.1. If you do not have a USB drive port to use with your computer/laptop, or USB port is malfunctioning, or one of your USB drives is malfunctioning, then additional configuration is required by the student to simulate the use of USB drives to export the required exam data throughout the course.

8.2. Refer to the ***2024 - Configuring VM Shared Folder in the host computer and accessed by the VM.pdf*** after the VB-VM installation has been completed.

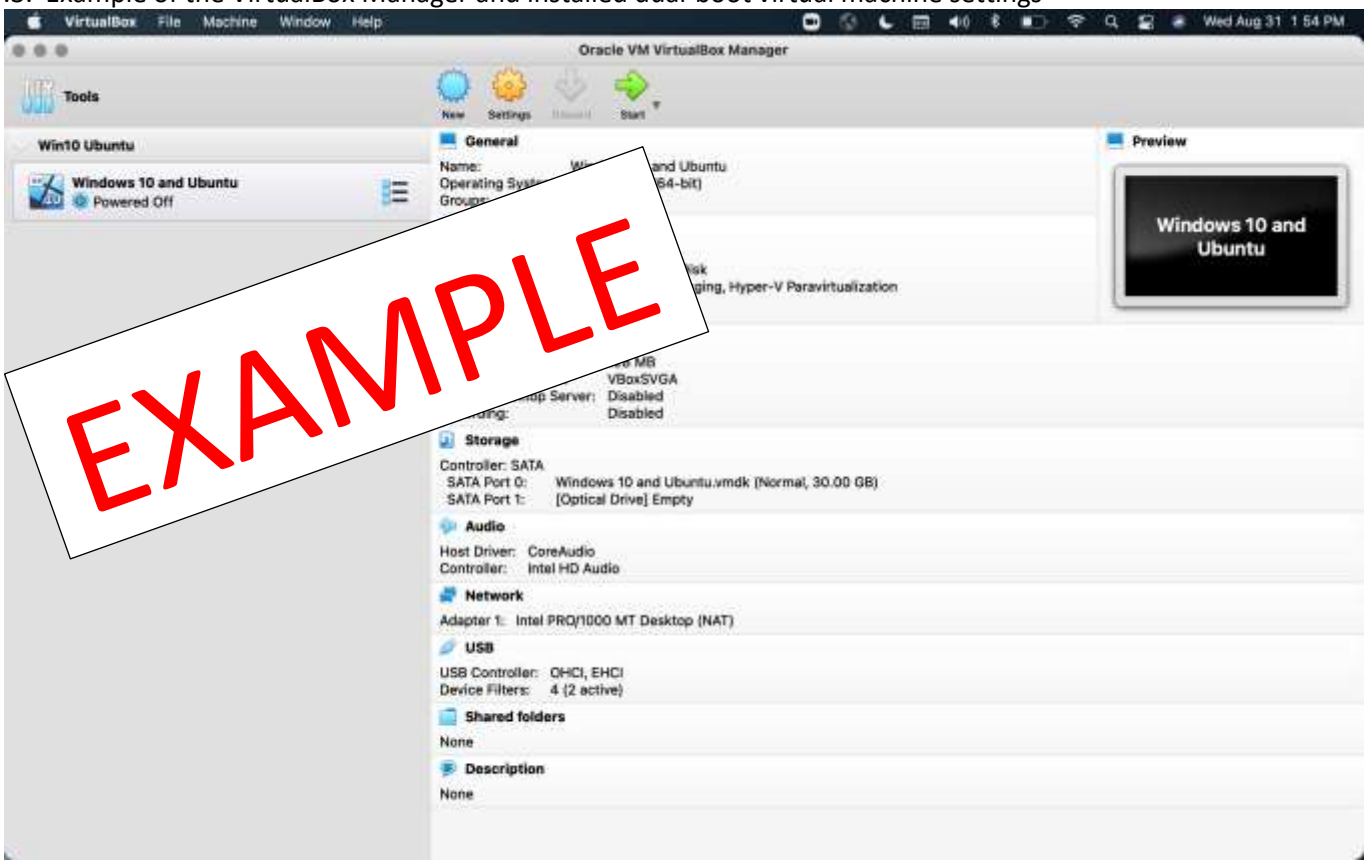
9. Appendix

9.1. Example of Windows login screen for installed VB-VM screenshots and their content.

9.2. Example of Linux login screen for installed VB-VM



9.3. Example of the VirtualBox Manager and installed dual-boot virtual machine settings



9.4. Ubuntu dual-bootloader menu (Ubuntu and Windows Recovery Environment)